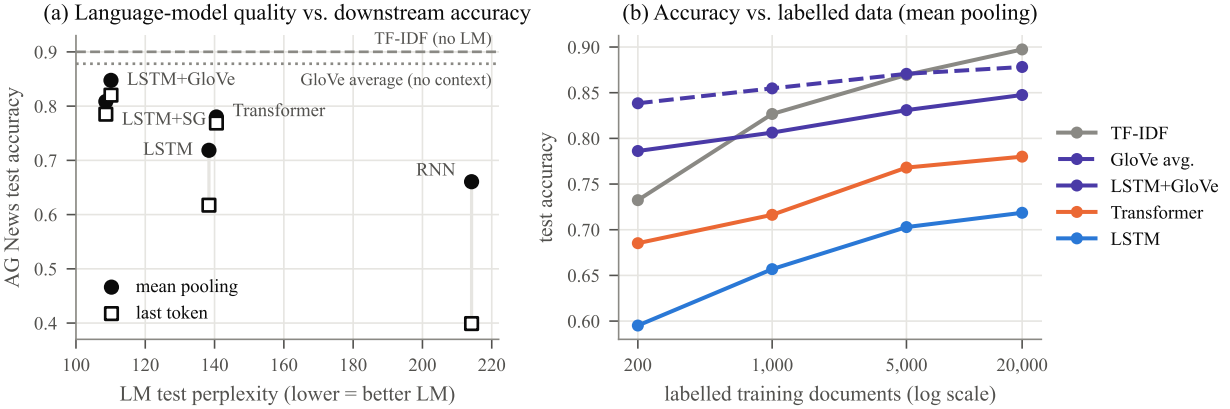
(a) Language-model quality vs. downstream accuracy
TF-IDF (no LM)
GloVe average (no context)
LSTM+GloVe
LSTM+SG
Transformer
LSTM
RNN
mean pooling
last token
AG News test accuracy
LM test perplexity (lower = better LM)
0.9
0.8
0.7
0.6
0.5
0.4
100
120
140
160
180
200
220
(b) Accuracy vs. labelled data (mean pooling)
test accuracy
labelled training documents (log scale)
0.90
0.85
0.80
0.75
0.70
0.65
0.60
200
1,000
5,000
20,000
TF-IDF
GloVe avg.
LSTM+GloVe
Transformer
LSTM